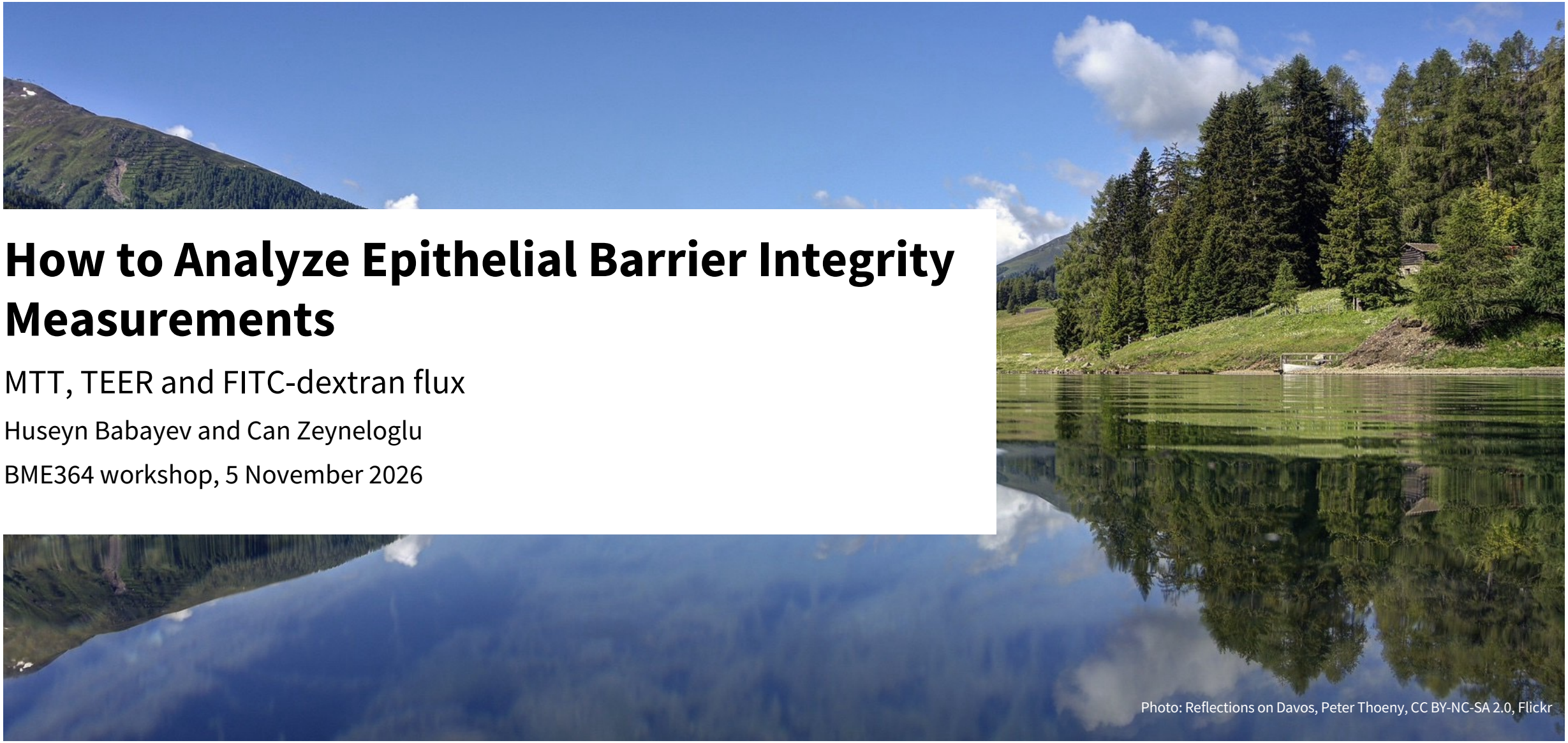
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How to Analyze Epithelial Barrier Integrity Measurements

MTT, TEER and FITC-dextran flux

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BME364 workshop, 5 November 2026

Photo: Reflections on Davos, Peter Thoeny, CC BY-NC-SA 2.0, Flickr

The Group I experiment in Davos

0 h

- TEER baseline (24-well Transwell)
- P80 apical: 0.02, 0.1, 0.5 %
- Control: medium only

24 h

- TEER
- FITC-dextran 4 kDa apical (0.5 mg/mL)

48 h

- TEER
- FD4 sampled apical + basolateral
- MTT (96-well): P80 1 → 0.03125 %

Four topics, one repository

01
Statistics
`01_statistics.Rmd`

02
MTT viability
`02_mtt.Rmd`

03
TEER
`03_teer.Rmd`

04
FITC-dextran flux
`04_flux.Rmd`

One experiment: n = inserts (TEER, flux; 4 per condition) or wells (MTT; 6 per concentration, 12 controls). Four groups → ANOVA + Dunnett vs control.

Parametric and nonparametric statistics

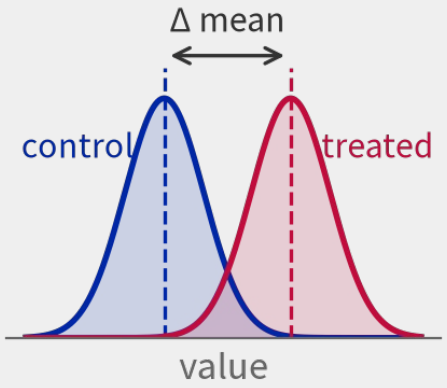
Theory, data structure, algorithm, implementation

01

Two families of tests, one logic

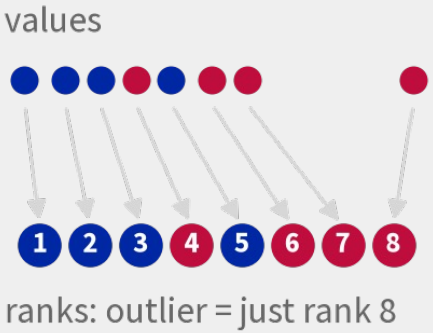
Parametric

- Normal residuals, equal variances, independence
- Compare **means**: t-test, ANOVA, mixed models
- More power when assumptions hold
- Effect: mean difference, 95 % CI



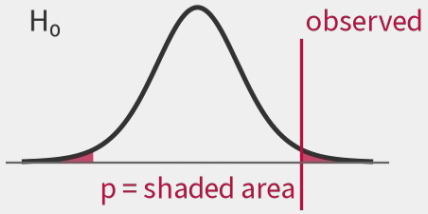
Nonparametric

- Values → **ranks**; no normality needed
- Compare **distributions**: Mann–Whitney, Kruskal–Wallis, Wilcoxon
- Robust to skew and outliers; less power at small n
- Effect: η^2_H, Hodges–Lehmann shift



Shared logic

H_0 → test statistic → p = P(data at least this extreme | H_0). α = 0.05 limits type I errors; power = 1 − β limits type II.



n in this course: technical replicates

One Caco-2 experiment → n = wells or inserts per condition. Conclusions hold for this experiment; reproducibility needs independent repeats (passages, days).

Structure decides the test

Wide: one column per group, hard to model

ctrl	P80
1.095	3.187
0.967	2.231

Long (tidy): one row per well, use this

well	group	value
W01	ctrl	1.095
W02	ctrl	0.967
W03	ctrl	1.492
W07	P80	3.187
W08	P80	2.231
W09	P80	2.045

Unit of analysis = n

One Caco-2 experiment: n = wells or inserts per condition. Images and cells inside a well are not n: average them first, or model them.

Independent vs paired

Different wells per condition → independent groups. Paired only when the same well or insert is measured twice, e.g. before and after, or over time.

Groups and factors

Two groups, or three and more; one factor, or two (treatment × time). This selects t-test vs ANOVA vs mixed model.

Decision algorithm: from design to test

1. **Unit and n**: the wells or inserts that were treated
2. **Pairing**: same well or insert measured again, or not
3. **Groups and factors**: 2 vs ≥ 3 groups; one or two factors
4. **Assumptions**: QQ plot / Shapiro–Wilk of residuals; Levene
5. **Test and post hoc**: from the table; correct for multiple tests
6. **Report**: effect with 95 % CI, exact p, n, test name

Design	Parametric	Nonparametric
2 independent	Welch t-test	Mann–Whitney U
2 paired	Paired t-test	Wilcoxon signed-rank
≥ 3 independent	One-way / Welch ANOVA	Kruskal–Wallis
≥ 3 matched	RM ANOVA / mixed model	Friedman
Post hoc	Tukey, Dunnett, Games–Howell	Dunn (Holm)
Association	Pearson r	Spearman ρ

Small n warning

With 3 vs 3 the smallest two-sided Mann–Whitney p is 0.10. Normality tests have little power at n ≤ 5; use QQ plots and what you know about the assay.

Statistics using R, chunk by chunk


```R
library(tidyverse); library(rstatix)
d <- read_csv("data/stats_example.csv")     # group, value

d %>% group_by(group) %>% shapiro_test(value)  # normal?
d %>% levene_test(value ~ group)               # equal SD?

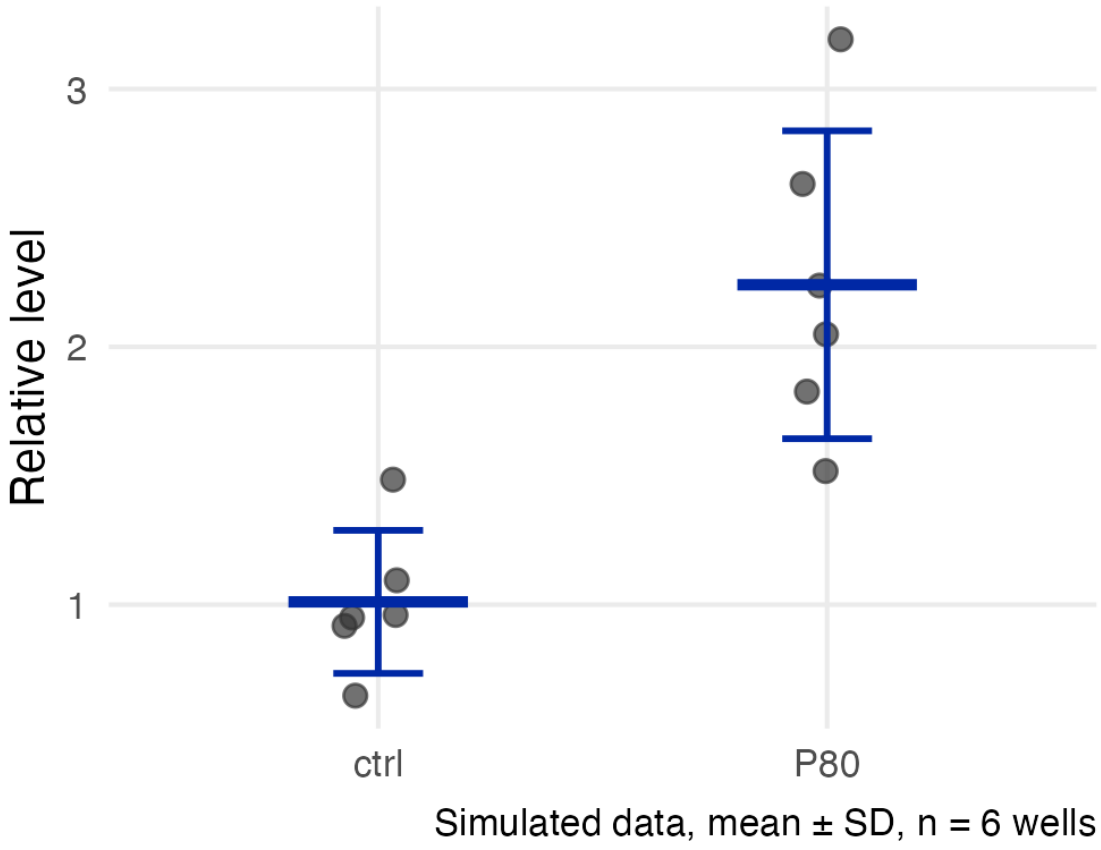
d %>% t_test(value ~ group, detailed = TRUE)  # Welch
d %>% t_test(value ~ group, var.equal = TRUE) # Student

d %>% wilcox_test(value ~ group)              # Mann-Whitney U

d %>% mutate(lv = log10(value)) %>% t_test(lv ~ group)
```

1. **Load**
   One row per well: group and value.
2. **Check the assumptions**
   Shapiro–Wilk per group, Levene across groups, plus QQ plots.
3. **Parametric**
   Welch t-test by default; Student only when the SDs are equal.
4. **Nonparametric**
   Mann–Whitney U on the ranks of all 12 wells.
5. **Skewed read-out**
   Log-transform, then the t-test again.

Result: both families agree



Test	Result
Shapiro–Wilk (ctrl / P80)	p = 0.49 / 0.91
Levene	p = 0.15
Welch t-test	t(7.1) = −4.58, p = 0.002
Mann–Whitney	U = 0, p = 0.002

P80 vs ctrl: higher

Difference 1.23 (95 % CI 0.60–1.86), log scale: p < 0.001

Ranks need replicates

U = 0: every P80 well above every control well. With 3 vs 3 the same pattern gives only p = 0.10.

MTT viability assay

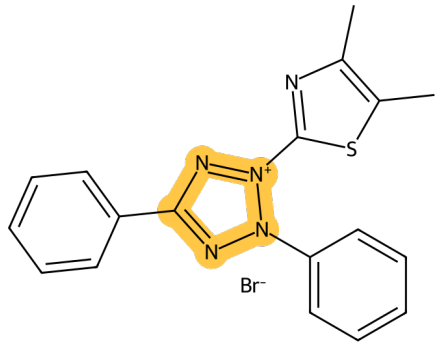
Theory, data structure, algorithm, implementation

02

MTT reports metabolic activity, not cell number

MTT

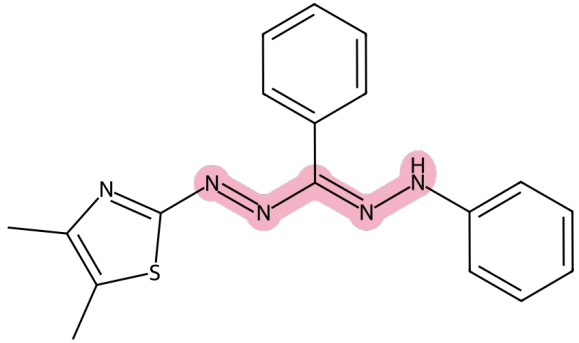
yellow, water-soluble tetrazolium salt



NAD(P)H oxidoreductases of living cells

Formazan

purple, insoluble crystals



dissolve in DMSO

Absorbance

540 nm, formazan dissolved in DMSO



more living cells → more formazan → darker purple → higher A540

Beer–Lambert

$A = \varepsilon \times c \times l$: absorbance is proportional to formazan, so to active cells (within the linear range of cell number).

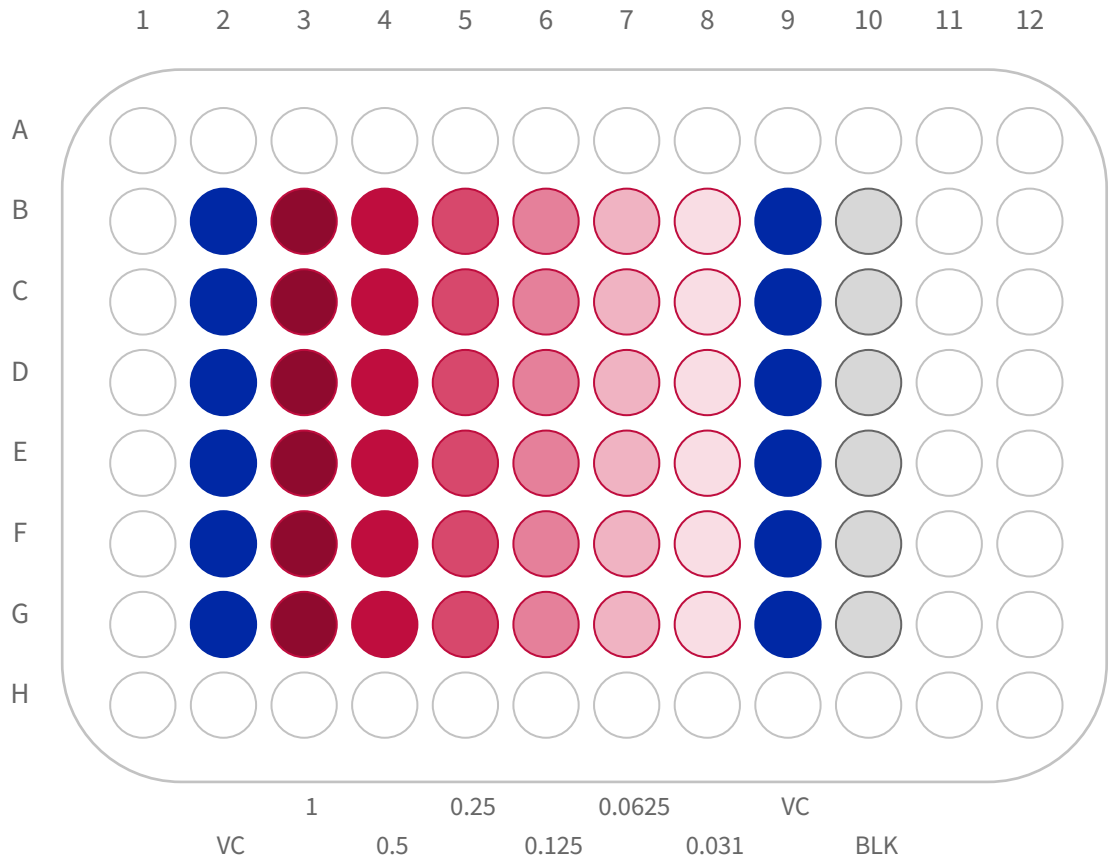
Activity, not number

Fewer cells and slower metabolism look the same. Confirm with a second read-out: LDH release, live/dead staining, cell count.

Interference

Reducing compounds (polyphenols, antioxidants) make formazan without cells: add a cell-free compound + MTT control.

Plate map becomes a long table



Cols 2 + 9 vehicle control (VC); **cols 3–8** P80 %; **col 10** blank
Rows B–G 6 replicate wells; 10,000 cells/well

well	condition	p80_pct	od540
B2	ctrl	0	0.728
B3	P80-1	1	0.095
B8	P80-0.03125	0.03125	0.702
B9	ctrl	0	0.683
B10	blank		0.094

data/mtt_raw.csv: 54 rows = 9 columns × 6 wells, one plate

One row per well

Well, condition and concentration travel with every reading; nothing depends on positions.

Unit = well

One plate, one experiment → n = 6 wells per concentration, 12 controls.

From optical density to a dose-response curve

$$\text{Viability (\%)} = (OD_{well} - \text{mean } OD_{blank}) / (\text{mean } OD_{control} - \text{mean } OD_{blank}) \times 100$$

1. **Blank**: mean of the medium + MTT + DMSO wells (no cells)
2. **Normalise**: (OD540 − blank) ÷ mean of the 12 control wells, × 100
3. **Check the plate**: control column 2 vs 9 should agree
4. **Assumptions**: residual QQ plot / Shapiro–Wilk; Levene
5. **Test**: ANOVA + Dunnett vs control; unequal SDs: Welch + Games–Howell
6. **Curve**: 4-parameter log-logistic fit → IC50 with 95 % CI

Why Dunnett, not 6 t-tests

Six comparisons with the same control: six t-tests at 0.05 inflate false positives; Dunnett keeps the family-wise error at 5 %.

Dead wells have almost no spread

Dunnett pools one SD over all groups. When SDs differ: Welch ANOVA + Games–Howell.

MTT analysis using R, chunk by chunk


```R
1 mtt   <- read_csv("data/mtt_raw.csv")
  blank <- mean(mtt$od540[mtt$condition == "blank"])

2 wells <- mtt %>% filter(condition != "blank") %>%
    mutate(od = od540 - blank,
           viability = 100 * od / mean(od[p80_pct == 0]),
           dose = factor(p80_pct))          # "0" = control

3 wells %>% filter(p80_pct == 0) %>%                # col 2 vs 9
    t_test(viability ~ column)
  wells %>% levene_test(viability ~ dose)           # equal SD?

4 fit <- aov(viability ~ dose, data = wells)
  emmeans(fit, trt.vs.ctrl ~ dose)                 # Dunnett
  wells %>% games_howell_test(viability ~ dose)     # unequal SD

5 curve <- drc::drm(viability ~ p80_pct, data = wells,
                    fct = drc::LL.4())
  drc::ED(curve, 50, type = "absolute", interval = "delta")
```

1 Blank

Medium + MTT + DMSO without cells.

2 % of control

Each well relative to the control mean.

3 Check plate and spread

Columns 2 vs 9: p = 0.75. SD 6–8 % alive, 1 % dead: Levene p = 0.01.

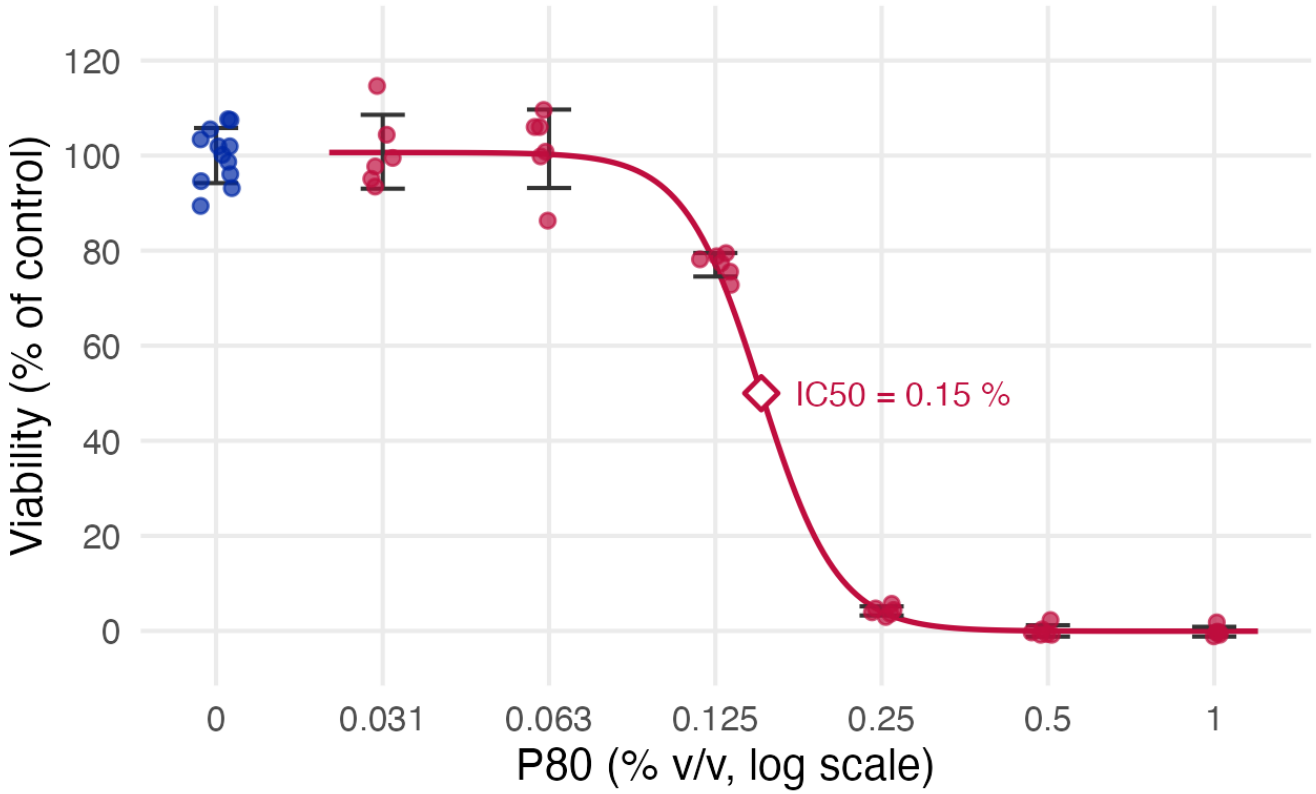
4 Each dose vs control

Both: ≥ 0.125 % lower ($p < 0.001$); 0.03125 and 0.0625 % not.

5 IC50

4-parameter log-logistic fit; 50 % viability.

Result: MTT dose response



Simulated data, points = wells (n = 6; control 12); mean ± SD; line = 4-parameter fit

Mean ± SD, n = 6 wells: 0.031 % 101 ± 8; 0.0625 % 101 ± 8; 0.125 % 77 ± 2; 0.25 % 4 ± 1; 0.5 % 0 ± 1; 1 % 0 ± 1

0.15 %

IC50 (95 % CI 0.14–0.16 %), steep 4-parameter fit

≥ 0.125 %

lower than control (Games–Howell); from 0.25 % almost no living cells

100, 94, 0 %

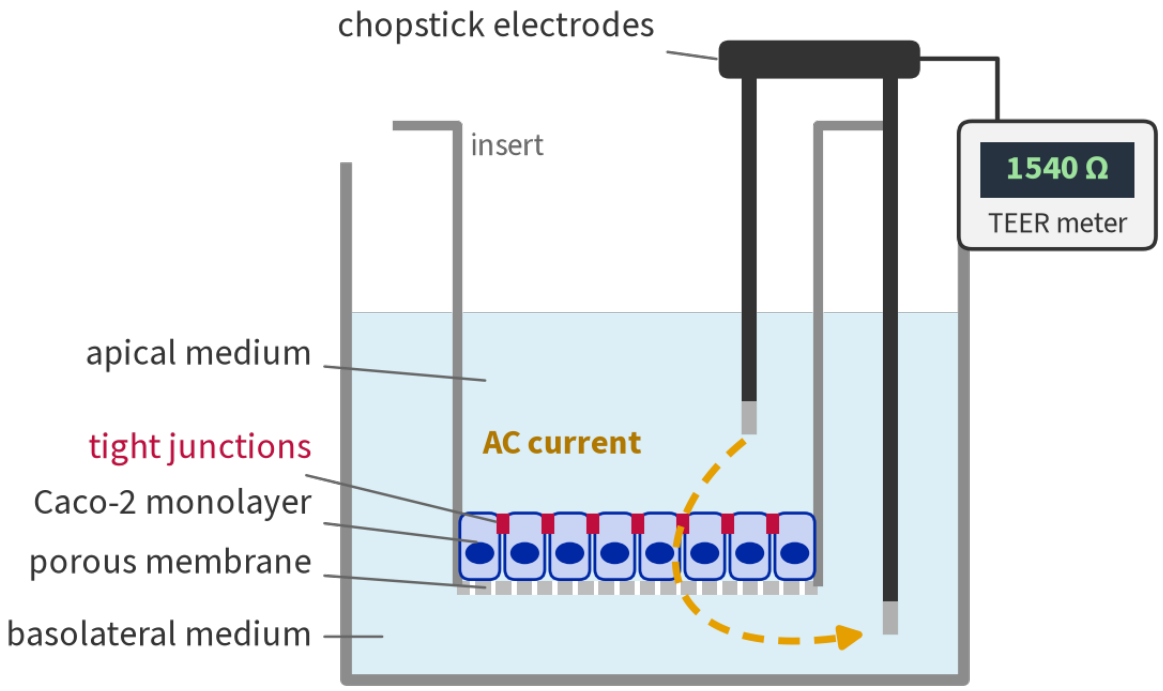
predicted viability at the barrier doses 0.02, 0.1, 0.5 %

Transepithelial electrical resistance (TEER)

Theory, data structure, algorithm, implementation

03

TEER is Ohm’s law across a cell layer



$R_{measured} = R_{blank}$ (medium + filter) + R_{cells}

R_{cells}: paracellular (tight junctions) in parallel with transcellular

Ohm’s law

R = V / I. A small AC current (≈ 12.5 Hz square wave) avoids electrode polarisation and harm to the cells.

Junctional ion flow

Mainly ion flow through tight junctions; it falls when junctions open.

Area-corrected: Ω cm²

Resistance is inversely proportional to area: multiplying by the membrane area makes inserts comparable.

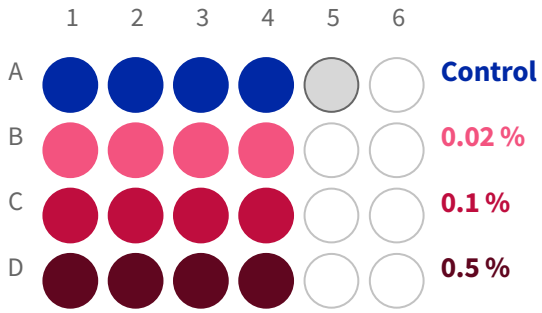
Sensitive to

Temperature (R falls as medium warms), electrode position and depth, medium changes.

Readings × inserts × time points

insert	condition	time_h	reading	r_ohm
A1	ctrl	0	2	1533
A1	ctrl	0	3	1539
A1	ctrl	24	2	1554
A1	ctrl	24	3	1547
A5	blank	0	2	180
A5	blank	0	3	184
…	…	…	…	…

data/teer_raw.csv: 102 rows = 17 inserts × 3 times × 2 readings



A5 = empty insert (blank), 24-well Transwell plate

Membrane area: Corning Transwell

Plate	Insert Ø	Area
24-well	6.5 mm	**0.33 cm²**
12-well	12 mm	1.12 cm²
6-well	24 mm	4.67 cm²

Technical level

2nd and 3rd reading per insert and time → average.

Blank

One empty insert (A5) with medium, read at every time point.

Repeated measures

Each insert is read at 0, 24 and 48 h: time points are not independent.

Independent inserts

4 conditions × 4 inserts on one plate: n = 4.

From ohms to Ω cm² and % of baseline

$$\text{TEER}\ (\Omega\ \text{cm}^2) = (R_{\text{insert}} - R_{\text{blank}}) \times A \qquad A = 0.33\ \text{cm}^2\ \text{(24-well insert)}$$

1. **Average readings**: mean of the 2nd and 3rd reading
2. **Subtract blank**: empty insert, same time point
3. **× area**: Ω → Ω cm² (0.33 cm²)
4. **% of baseline**: divide by the insert's own 0 h value
5. **End-point** at 48 h: ANOVA + Dunnett vs control (unequal SDs: Games–Howell)
6. **Time course**: mixed model, condition × time, insert as random effect

Worked example (SIAF data, Ogulur et al., Allergy 2023)

(1548 − 175 Ω) × 0.33 cm² = 1373 Ω × 0.33 cm² = 453 Ω cm²

Each insert is its own baseline

Inserts start at different TEER (here 370–550 Ω cm²): each insert becomes its own reference before P80 is added.

TEER analysis using R, chunk by chunk

R

```r
# 1
teer <- read_csv("data/teer_raw.csv") %>%
  group_by(insert, condition, time_h) %>%
  summarise(r_ohm = mean(r_ohm), .groups = "drop")

# 2
blank <- teer %>% filter(condition == "blank") %>%
  select(time_h, r_blank = r_ohm)
teer <- teer %>% filter(condition != "blank") %>%
  left_join(blank, by = "time_h") %>%
  mutate(teer = (r_ohm - r_blank) * 0.33)          # ohm * cm2

# 3
teer <- teer %>% group_by(insert) %>%
  mutate(pct = 100 * teer / teer[time_h == 0])

# 4
end48 <- teer %>% filter(time_h == 48)
end48 %>% levene_test(pct ~ condition)          # p = 0.002
end48 %>% games_howell_test(pct ~ condition)    # vs ctrl

# 5
fit <- lmer(teer ~ condition * factor(time_h) + (1 | insert),
            data = teer)
```

1 Average the readings

2nd and 3rd reading → one value per insert and time.

2 Blank, × area

Empty insert at the same time; × 0.33 cm^2 → Ω cm^2.

3 % of baseline

Each insert is its own reference (0 h).

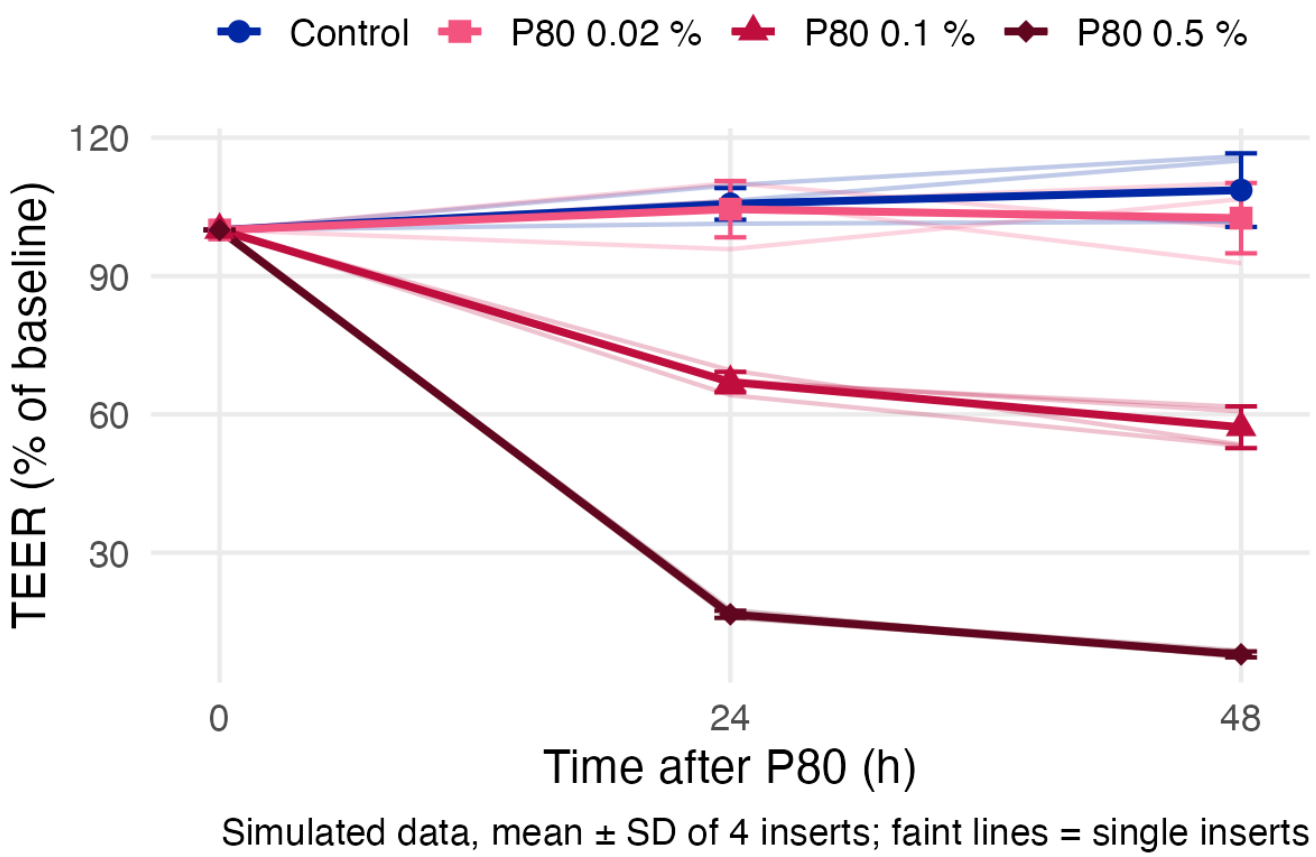
4 End-point at 48 h

Each P80 concentration vs control; SDs differ → Games–Howell.

5 Whole time course

Mixed model: inserts are read repeatedly.

Result: TEER over 48 h



Simulated data, mean ± SD of 4 inserts; faint lines = single inserts

57 %

0.1 % at 48 h (control 109 %), Games–Howell $p < 0.001$; 0.02 %: 103 %, $p = 0.70$

8 %

0.5 % at 48 h; the MTT shows no living cells at 0.5 %

F = 83

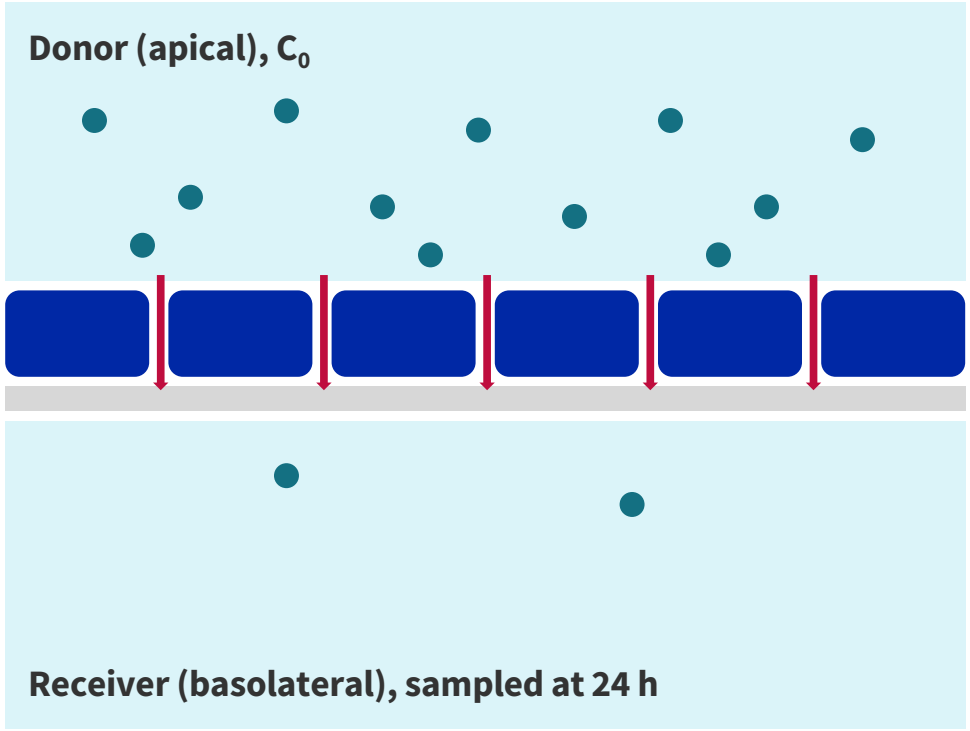
condition × time, mixed model, $F(6, 24)$; $p < 0.001$

Paracellular flux (FITC-dextran 4 kDa)

Theory, data structure, algorithm, implementation

04

Flux measures tracer transport across the barrier



Gaps between cells = paracellular route (red arrows).

After 24 h both sides are sampled: 100 µL in duplicate wells.

Fick's first law

$J = (dQ/dt) / A = P_{app} \times C_0$. One end-point: $dQ/dt \approx Q / t$ while the receiver stays below ~10 % of the donor.

Apparent permeability

P_{app} in cm/s normalises the rate for area and donor concentration, comparable across inserts and labs.

Tracer size selects the pathway

Na-fluorescein (376 Da), Lucifer yellow (457 Da): small solutes. FITC-dextran 4 kDa (FD4) and larger: leak pathway.

Read together with TEER

TEER ↓ with unchanged FD4 flux → pore (ion) pathway; FD4 flux ↑ → leak pathway or lesions.

Two tables and a set of constants

Standards (duplicates)

conc_ug_ml	rfu
0	52
0	47
0.39	409
0.39	427
0.78	786
…	…
12.5	12 238
25	23 401
25	23 821

Samples (64 rows = 16 inserts × 2 sides × 2)

insert	side	dilution	rfu
A1	basolateral	1	1087
A1	basolateral	1	1103
A1	apical	50	9852
A1	apical	50	9716
…	…	…	…

Stay inside the standards

- Range 0.39–25 µg/mL: apical samples (~500 µg/mL) diluted 1 : 50
- Duplicate wells → average; they are not n
- Units in column names: conc_ug_ml, time in s

Constants

- $A = 0.33\ cm^2$
- $C_0 = 500\ µg/mL$ (0.5 mg/mL)
- $V_{api} = 0.3\ mL$
- $V_{baso} = 0.6\ mL$
- $t = 24\ h = 86\ 400\ s$

Same plate, same medium

Standards in cDMEM next to the samples; blank = medium only.

From fluorescence to apparent permeability

$$P_{app} = (C_{baso} \times V_{baso}) / (t \times A \times C_0) \qquad \mu g \div (s \times cm^2 \times \mu g/cm^3) = cm/s$$

1. **Standard curve**: linear fit RFU = a + b × C over 0.39–25 µg/mL; R^2
2. **Concentration**: mean of duplicates; C = (RFU − a) / b × dilution
3. **Amount crossed**: $C_{baso} \times V_{baso}$; % of the dose ($C_0 \times V_{api}$)
4. **Recovery**: apical + basolateral ≈ 100 % of the dose
5. **P_{app}**: end-point, in cm/s
6. **Test**: $\log_{10} P_{app}$: ANOVA + Dunnett vs control; Spearman with TEER

498 / 517 nm are settings, not two readings

Excitation 498 nm and emission 517 nm define one fluorescence reading; concentration comes from the standard curve, not from a subtraction.

Sink conditions

Keep the transported fraction below ~10 % of the dose so the gradient stays constant (here ≤ 9 %: close to the limit at 0.5 % P80).

Flux analysis using R, chunk by chunk


```R
1 cal <- lm(rfu ~ conc_ug_ml, data = std)            # R2 = 0.9996

2 conc <- smp %>% group_by(insert, condition, side, dilution) %>%
    summarise(rfu = mean(rfu), .groups = "drop") %>%  # duplicates
    mutate(conc = (rfu - coef(cal)[1]) / coef(cal)[2] * dilution)

3 papp <- conc %>% select(insert, condition, side, conc) %>%
    pivot_wider(names_from = side, values_from = conc) %>%
    mutate(crossed = basolateral * 0.6,              # ug
           papp = crossed / (86400 * 0.33 * 500))    # cm/s

4 papp <- papp %>% mutate(log_papp = log10(papp))
  fit  <- aov(log_papp ~ condition, data = papp)
  emmeans(fit, trt.vs.ctrl ~ condition)              # Dunnett

5 both %>% cor_test(teer, papp, method = "spearman")  # TEER 48 h
```

1 Standard curve

7 standards, 25 → 0.39 µg/mL.

2 RFU → µg/mL

Mean of duplicates; × dilution (apical 1:50).

3 Amount → P_{app}

$C_{baso} \times V_{baso} / (t \times A \times C_0)$.

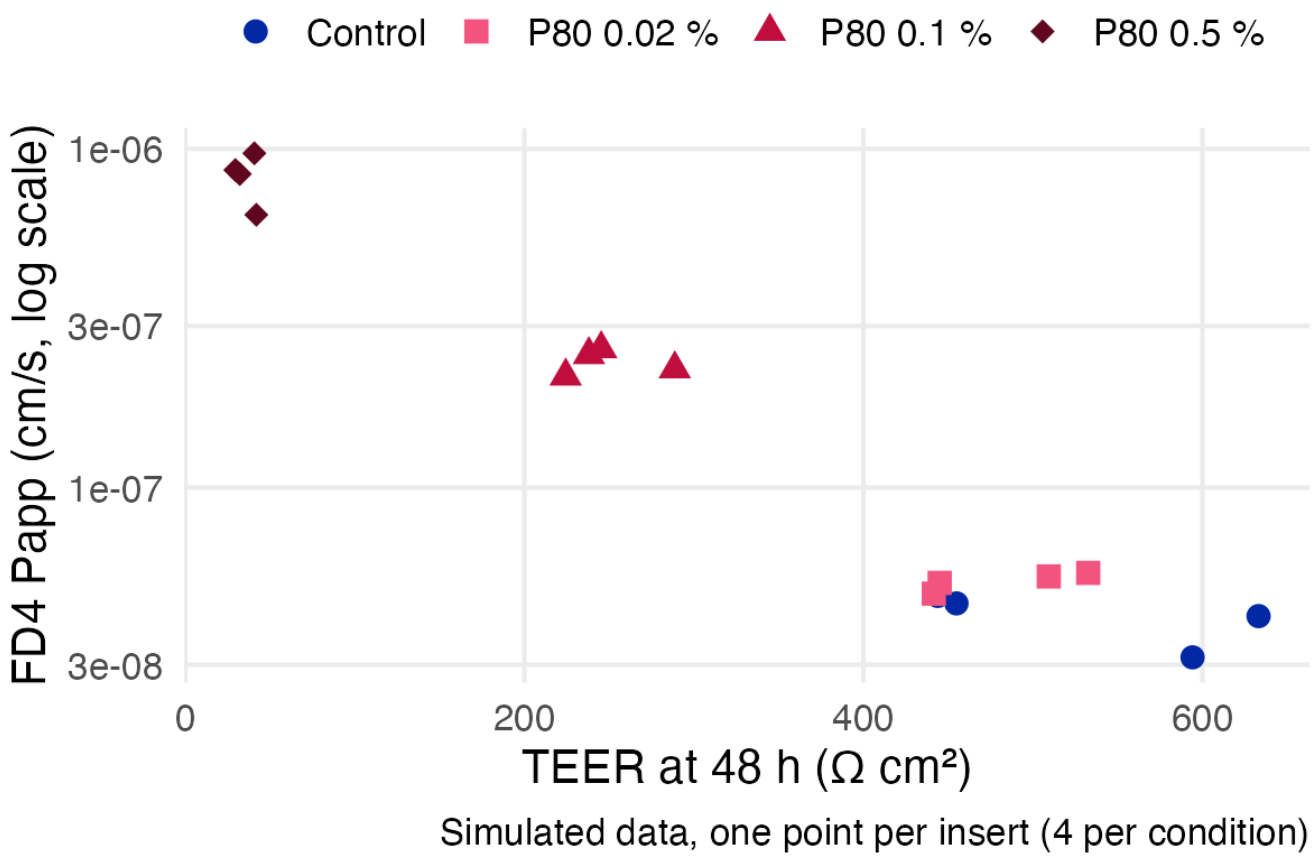
4 Test on log scale

P_{app} spans orders of magnitude; $10^{estimate}$ = fold change.

5 Relate to TEER

Spearman ρ across the 16 inserts.

Result: permeability and TEER



1.3, 5.7, 20 ×

P_{app} at 0.02, 0.1, 0.5 % vs control (geometric means)

p < 0.001

0.1 and 0.5 %, Dunnett on $\log_{10} P_{app}$; 0.02 %: p = 0.07

ρ = −0.89

TEER 48 h vs P_{app}, 16 inserts, p < 0.001

Before you report barrier data: a checklist

1. n = wells or inserts of one experiment (say so in every legend)
2. MTT: blank subtracted, both control columns agree, control = 100 %, IC50
3. ≥ 3 groups: ANOVA + Dunnett vs control; Games–Howell when SDs differ
4. Exact p-values, test named, effect with 95 % CI
5. TEER in Ω cm^2: blank subtracted, × 0.33 cm^2, % of baseline
6. Time course with a mixed model: inserts are read repeatedly
7. FD4: standards bracket every sample; dilutions and recovery stated
8. P_{app} with tracer, time, area and C_0; read together with TEER

Run everything yourself

```
BME364-G1-share/
├── README.md          start here
├── R/                 mac_linux/, windows/ (.Rmd)
├── data/              example data (.csv)
├── example_images/    4 real IF fields (.tif)
├── fiji/              measure_zo1.ijm
├── figures/           plots, written by R
├── results/           tables, written by R
├── manuals/           lab manuals (.docx)
└── slides/            both decks (.pptx)
```

R

```r
# R: R/mac_linux/ or R/windows/
# once: Source 00_install_packages.R
# knit in RStudio, in this order:
#   01_statistics.Rmd, 02_mtt.Rmd,
#   03_teer.Rmd, 04_flux.Rmd (needs 03)
```

Your own data from Davos

- R packages: rmarkdown, tidyverse, rstatix, lme4, lmerTest, emmeans, drc
- Replace the files in data/ with your own exports and keep the column names; the .Rmd files then reproduce every step of this workshop.